Near-term Quantum Computing — lecture notes (test)

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Preface

These notes accompany the lectures of *Near-term Quantum Computing* (UvA, 2026-2027). This is a **test build** for trying out GitHub Pages: it contains one section of Lecture 1 only.

Version: 2026-10-15. Changes after release are listed on a changelog page.

1 Native gates and connectivity

A circuit as written in an SDK is not what runs on a device. The hardware implements a small set of *native gates*, each a calibrated control pulse, and only between the pairs of qubits that are physically coupled. This section builds the one identity we need to read native gate sets, and then looks at what connectivity costs.

1.1 Pauli rotations

Definition 1.1 (Pauli string)**.** A *Pauli string* on n qubits is a tensor product $P = P_1 \otimes P_2 \otimes \cdots \otimes P_n$ with each $P_j \in \{I, X, Y, Z\}$. We often drop the tensor product signs and write, for example, ZZ or XIY.

Every Pauli string is Hermitian and squares to the identity, $P^2 = I$, because each single-qubit factor does. This one property is all we need for the following identity.

Proposition 1.1 (Pauli rotations)**.** *Let P be a Hermitian operator with $P^2 = I$, for example a Pauli string. Then for all real θ*

$$e^{-i\theta P/2} = \cos\frac{\theta}{2}\, I - i \sin\frac{\theta}{2}\, P. \tag{1.1}$$

Proof. Expand the exponential as a power series. Since $P^2 = I$, even powers give $P^{2m} = I$ and odd powers give $P^{2m+1} = P$, so

$$e^{-i\theta P/2} = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} \left(\frac{\theta}{2}\right)^{2m} I \;-\; i \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \left(\frac{\theta}{2}\right)^{2m+1} P,$$

and the two series are those of $\cos(\theta/2)$ and $\sin(\theta/2)$. □

Example 1.1 (Single- and two-qubit rotations)**.** Applying Equation 1.1 with $P = X$, Z and $Z \otimes Z$ gives

$$R_X(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \tag{1.2}$$

$$R_{ZZ}(\theta) = e^{-i\theta Z\otimes Z/2} = \operatorname{diag}\!\left(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2}\right). \tag{1.3}$$

R_{ZZ} applies a phase that depends only on the parity of the two bits. That is why it can be built from two CNOTs and a single R_Z (Figure 1.1): the first CNOT writes the parity $a \oplus b$ into the second qubit, R_Z applies the phase, and the second CNOT undoes the first.

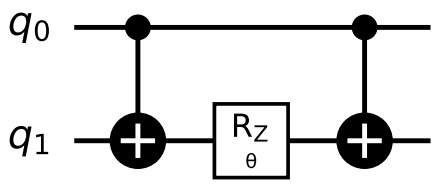


Figure 1.1: The decomposition $R_{ZZ}(\theta) = \mathrm{CNOT}\,(I \otimes R_Z(\theta))\,\mathrm{CNOT}$, drawn with Qiskit.

> **Where this returns**
>
> Equation 1.1 comes back twice later in the course: in Trotter circuits, where e^{-iHt} is split into Pauli rotations, and in the parameter-shift rule for gradients.

1.2 Native gate sets

Table 1.1 lists the native gates of some current platforms. Gate names and details change with each hardware generation; the point is the pattern. Every platform offers a continuous family of single-qubit rotations and one or two entangling gates.

Table 1.1: Native gates of some current devices. `TODO` verify against current device specifications.

Platform	Single-qubit	Two-qubit
IQM (superconducting)	$\mathrm{PRX}(\theta, \varphi)$	CZ
IBM (superconducting)	R_Z (virtual), $\sqrt{X}$, X	CZ or ECR
Google (superconducting)	PhasedXZ	CZ, $\sqrt{i\mathrm{SWAP}}$
Quantinuum (trapped ions)	$U_{1q}(\theta, \varphi)$, R_Z	$R_{ZZ}(\theta)$, any angle
IonQ (trapped ions)	GPi, GPi2	Mølmer-Sørensen

The IQM gate is a rotation about an axis in the xy-plane of the Bloch sphere,

$$\mathrm{PRX}(\theta, \varphi) = \exp\left[-i\frac{\theta}{2}\left(\cos\varphi\, X + \sin\varphi\, Y\right)\right]. \tag{1.4}$$

The operator in brackets squares to the identity, so Proposition 1.1 applies to it too.

1.2.1 Why R_Z is free

On superconducting devices, R_Z is usually a *virtual Z gate*: no pulse is applied at all. The reason is the conjugation identity

$$R_Z(\alpha)\,\mathrm{PRX}(\theta,\varphi)\,R_Z(\alpha)^\dagger = \mathrm{PRX}(\theta,\varphi+\alpha), \tag{1.5}$$

which follows from $R_Z(\alpha)XR_Z(\alpha)^\dagger = \cos\alpha\,X + \sin\alpha\,Y$. Using Equation 1.5, every R_Z can be moved later through the circuit, changing the phases φ of the drive pulses it passes. CZ commutes with R_Z, so nothing else changes. At the end, an R_Z right before a measurement in the computational basis has no effect and can be dropped.

1.3 Connectivity

A native two-qubit gate is available only between qubits that are physically coupled. The *coupling graph* records which pairs these are. On a superconducting chip it is fixed by fabrication: a square lattice, IBM's heavy-hex lattice, or IQM's star layout around a central resonator. In a trapped-ion chain all pairs can interact, at the price of slower gates.

A two-qubit gate between qubits that are not neighbours needs SWAP gates to bring them together, and each SWAP costs three CNOTs (Figure 1.2).

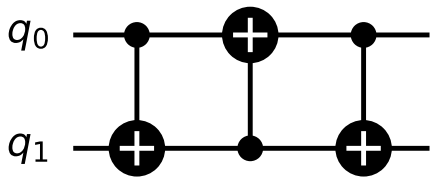


Figure 1.2: SWAP as three CNOTs with alternating control.

In practice we do not insert SWAPs by hand: *transpilation* maps a circuit onto the native gates and coupling graph of a device. We return to its cost with the algorithms; for now, remember that connectivity turns into extra two-qubit gates, and two-qubit gates are the main source of error (Preskill 2018).

Exercise 1.1 (Cost of distance)**.**

(a) Check Figure 1.2 on the four computational basis states.
(b) On a line of n qubits, how many CNOTs does a CNOT between the two end qubits cost, if the layout need not be restored afterwards? And if it must?

> **Try it**
>
> Tutorial 1 builds these circuits in Qiskit and transpiles them to a coupling graph: open the T1 notebook in Colab. For background on the circuit model, see Nielsen and Chuang (2010), chapter 4.

References

Nielsen, Michael A., and Isaac L. Chuang. 2010. *Quantum Computation and Quantum Information*. 10th anniversary. Cambridge University Press.

Preskill, John. 2018. "Quantum Computing in the NISQ Era and Beyond." *Quantum* 2: 79. https://doi.org/10.22331/q-2018-08-06-79.

A Glossary

coupling graph Graph whose vertices are qubits and whose edges are the pairs on which a native two-qubit gate is available.

native gate A gate the hardware implements directly with a calibrated control pulse. Every other gate must be compiled into native gates.

Pauli string A tensor product of single-qubit Paulis, P = P_1 … P_n with each P_j {I, X, Y, Z}. Hermitian, and squares to the identity.

transpilation Compilation of a circuit into the native gates and coupling graph of a device, including inserting SWAPs and simplifying.

virtual Z gate An R_Z rotation implemented by shifting the phase of all later drive pulses instead of applying a pulse. It takes no time and adds no error.